

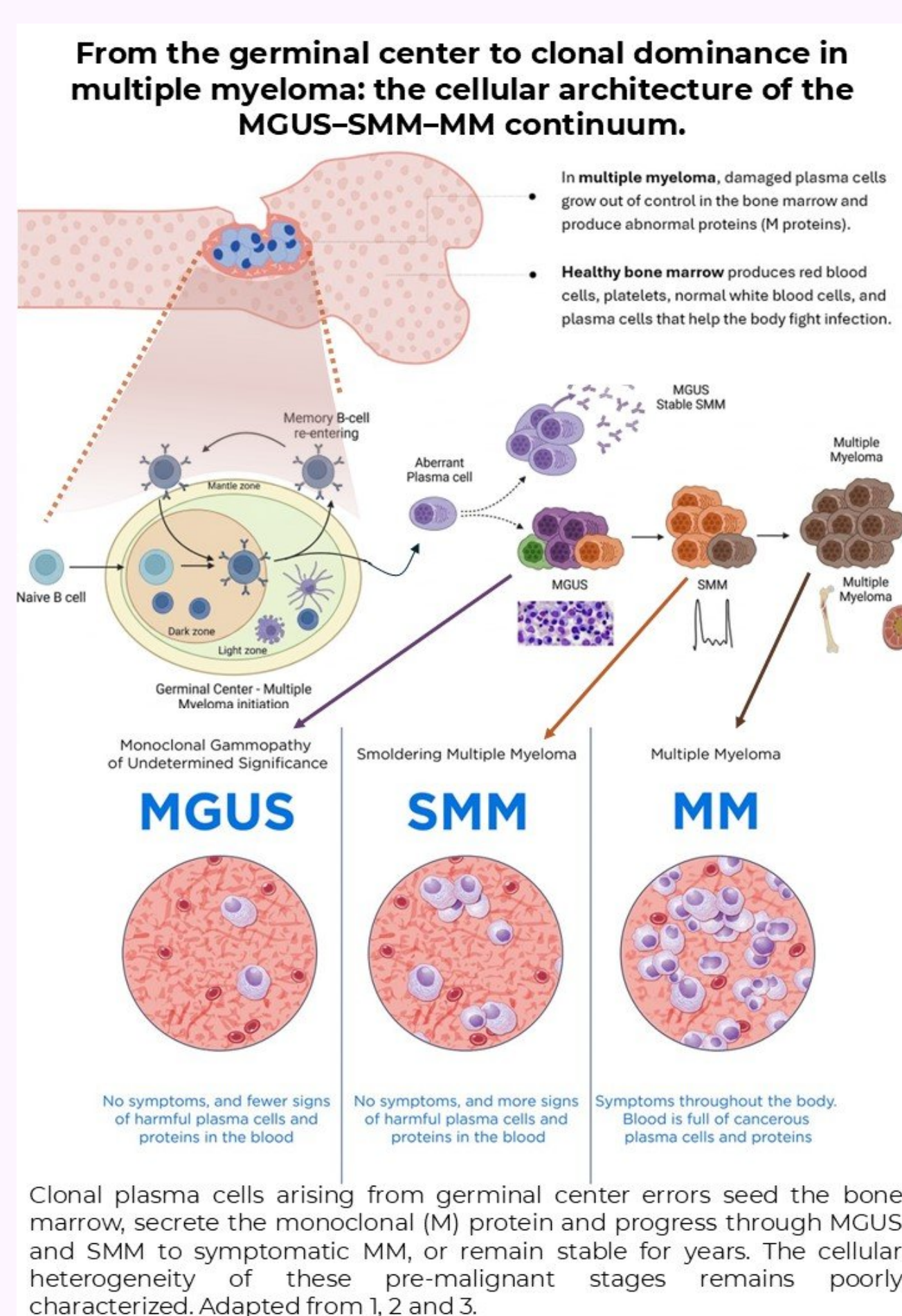
When Does Myeloma Really Begin? *Single-Cell Evidence That Transformation Precedes Diagnosis.*

Tiago Fernando Chaves^{1,2,*}, Juliane Heyde dos Santos^{1,2}, Taline Maria Fontana^{1,2}, Ricardo Henrique Döring^{1,2}, Bruna Terezinha Magnabosco Ferreira da Cruz², Juliana Dal-Ri Lindenau¹, Daniel Pacheco Bruschi², Yara Costa Netto Muniz¹.

¹Laboratório de Polimorfismos Genéticos (LAPOGE). Departamento de Biologia Celular, Embriologia e Genética, Universidade Federal de Santa Catarina (UFSC).;

²Laboratório de Citogenômica Evolutiva. Departamento de Genética, Universidade Federal do Paraná (UFPR). *E-mail: chaves.t.f@posgrad.ufsc.br

Introduction

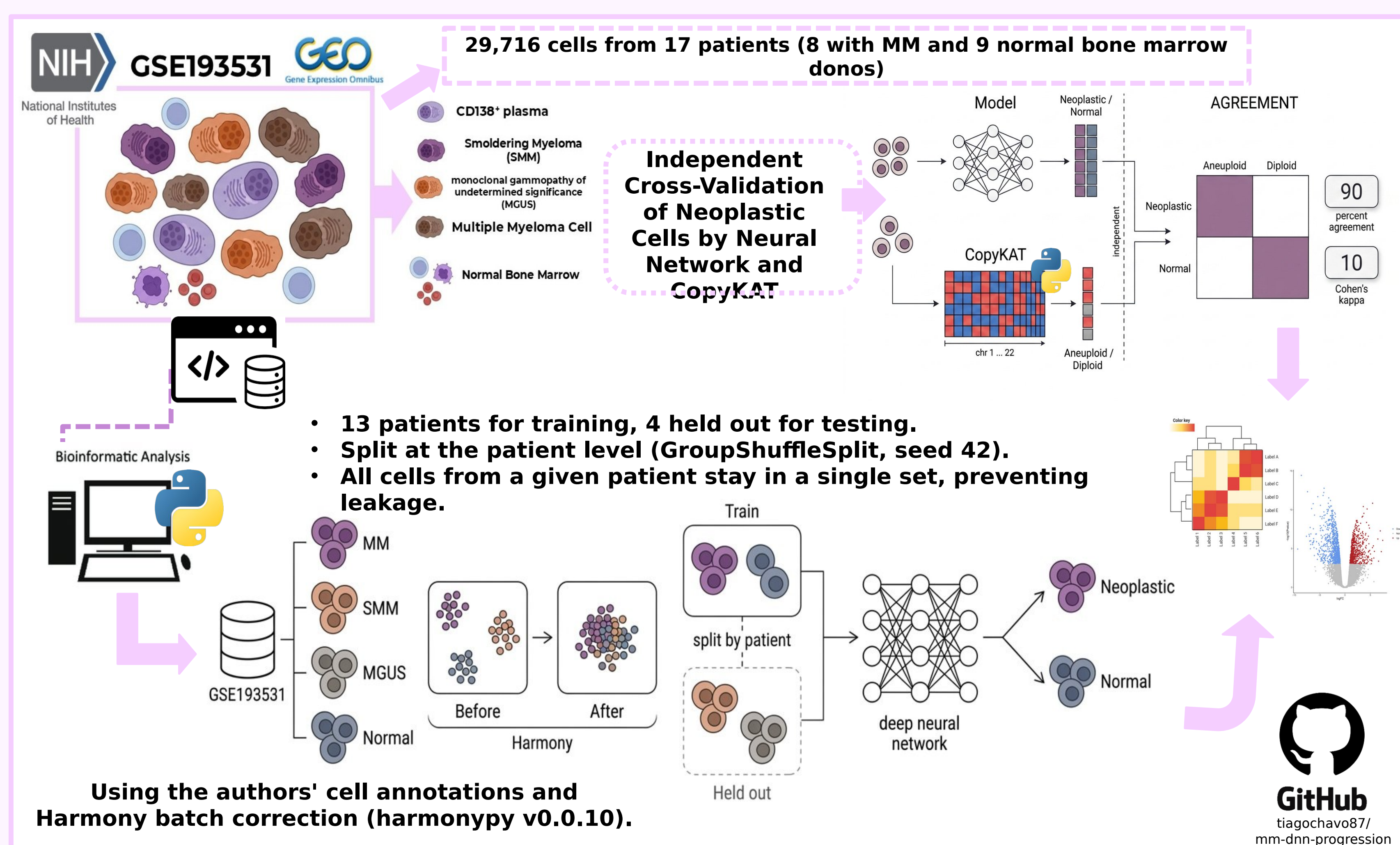


- Multiple myeloma (MM) is an incurable plasma cell malignancy.
- 188,000 new cases and 121,000 deaths worldwide in 2022.
- Cases projected to rise 70.8% by 2045.
- It is always preceded by MGUS and SMM, both asymptomatic.
- Progression to active disease is not predicted by current clinical markers.
- Who progresses, and when, remains unknown.

When Does Myeloma Really Begin?

We aimed to identify, at single-cell resolution, individual plasma cells within MGUS and SMM that already carry a myeloma-like transcriptional profile.

Methods



Results and Discussion

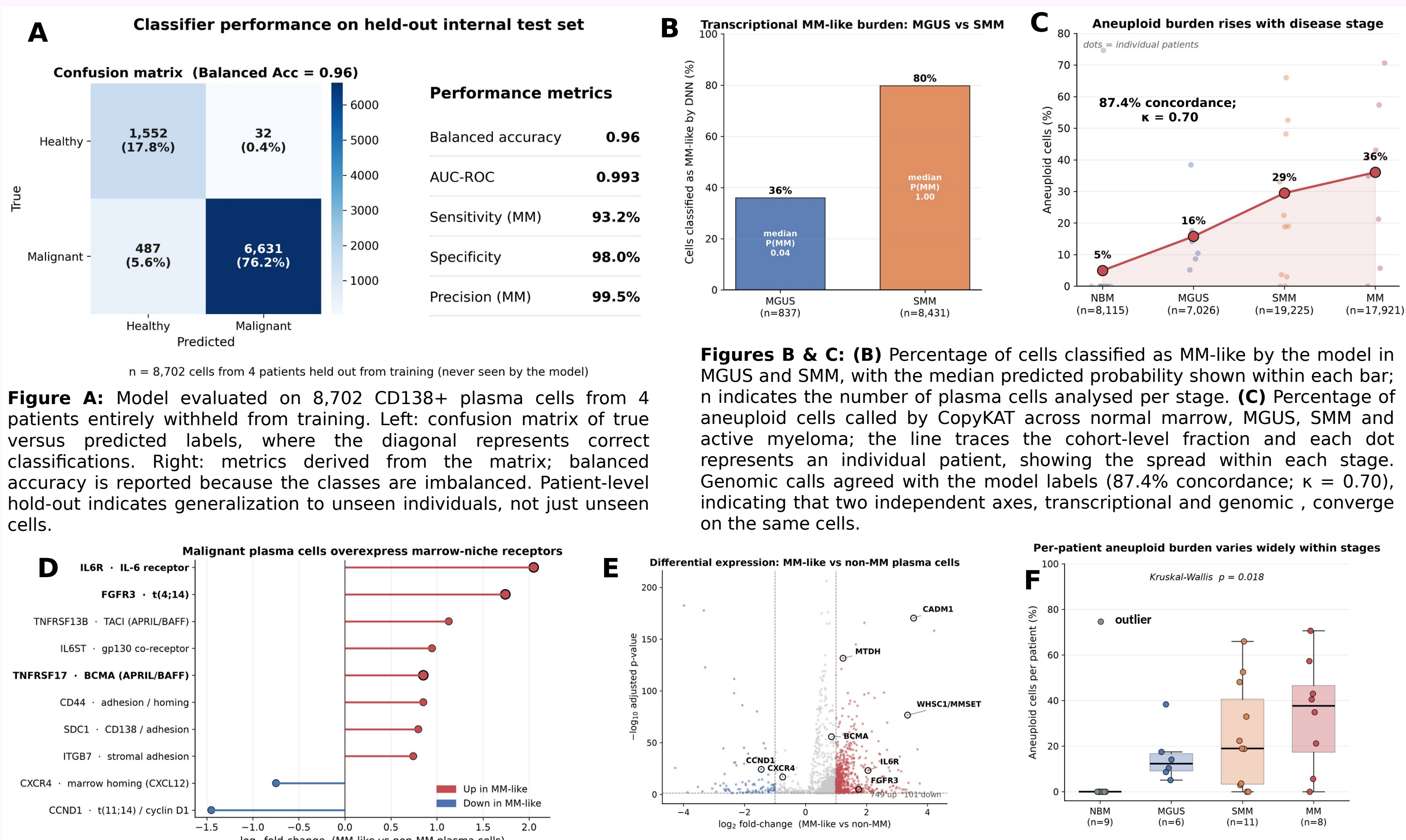


Figure A: Model evaluated on 8,702 CD138⁺ plasma cells from 4 patients entirely withheld from training (never seen by the model) versus predicted labels, where the diagonal represents correct classifications. Right: metrics derived from the matrix; balanced accuracy is reported because the classes are imbalanced. Patient-level hold-out indicates generalization to unseen individuals, not just unseen cells.

Figures B & C: (B) Percentage of cells classified as MM-like by the model in MGUS and SMM, with the median predicted probability shown within each bar; n indicates the number of plasma cells analysed per stage. (C) Percentage of aneuploid cells called by CopyKAT across normal marrow, MGUS, SMM and active myeloma; the line traces the cohort-level fraction and each dot represents an individual patient, showing the spread within each stage. Genomic calls agreed with the model labels (87.4% concordance; $\kappa = 0.70$), indicating that two independent axes, transcriptional and genomic, converge on the same cells.

Figures D, E & F: (D) $\log_2$ fold-change of marrow-niche receptors in MM-like versus non-MM plasma cells; red denotes up-regulation and blue down-regulation, with IL6R, FGFR3 and BCMA highlighted. (E) Differential expression between MM-like and non-MM plasma cells; dashed lines mark the significance thresholds ($|\log_2FC| \geq 1$, adjusted $p < 0.05$) and myeloma-associated genes are labelled. (F) Aneuploid cell fraction per patient across stages, where each dot is one patient and boxes show the median and interquartile range; burden varied widely within MGUS and SMM (Kruskal-Wallis $p = 0.018$), indicating that malignant load is a property of the individual rather than of the stage label.

- An active-MM signature is already present in MGUS and SMM;
- Two methods agree: Deep learning + aneuploidy;
- Malignant, druggable program (WHSC1/MMSET, FGFR3, IL6R, BCMA);
- Malignant burden is a single-cell trait that varies within stages.

Conclusion

Neoplastic, aneuploid plasma cells are already present before clinical progression of MM, and their per-patient burden differs markedly among MGUS and SMM individuals, a heterogeneity that may help refine risk stratification in these stages. Single-Cell Evidence That Transformation Precedes Diagnosis.

References

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